

实验 2 语法分析器-算术表达式

语法分析器分两部分，第一部分为算术表达式，第二部分为布尔表达式和控制语句。

要求

参考课本 4.4，实现递归下降语法分析器。

当语法分析器需要单词符号时，调用词法分析器获取单词符号。

文法

$expr \rightarrow term\ rest5$

$rest5 \rightarrow +term\ rest5 \mid -term\ rest5 \mid \epsilon$

$term \rightarrow unary\ rest6$

$rest6 \rightarrow *unary\ rest6 \mid /unary\ rest6 \mid \epsilon$

$unary \rightarrow factor$

$factor \rightarrow \mathbf{num}$

步骤

可将以上文法分解为小的文法分步完成。

第一步 包含乘、除的算术表达式

$term \rightarrow unary\ rest6$

$rest6 \rightarrow *unary\ rest6 \mid /unary\ rest6 \mid \epsilon$

$unary \rightarrow factor$

$factor \rightarrow \mathbf{num}$

输入：

5*2/3

输出（二选一）：

1) 按使用产生式过程

- (1) $term \rightarrow unary\ rest6$
- (2) $unary \rightarrow factor$
- (3) $factor \rightarrow num$
- (4) $rest6 \rightarrow *unary\ rest6$
- (5) $unary \rightarrow factor$
- (6) $factor \rightarrow num$
- (7) $rest6 \rightarrow /unary\ rest6$
- (8) $unary \rightarrow factor$
- (9) $factor \rightarrow num$
- (10) $rest6 \rightarrow \epsilon$

2) 按推导过程

- (1) $\text{term} \Rightarrow \text{unary rest6}$
- (2) $\Rightarrow \text{factor rest6}$
- (3) $\Rightarrow \text{num rest6}$
- (4) $\Rightarrow \text{num} * \text{unary rest6}$
- (5) $\Rightarrow \text{num} * \text{factor rest6}$
- (6) $\Rightarrow \text{num} * \text{num rest6}$
- (7) $\Rightarrow \text{num} * \text{num} / \text{unary rest6}$
- (8) $\Rightarrow \text{num} * \text{num} / \text{factor rest6}$
- (9) $\Rightarrow \text{num} * \text{num} / \text{num rest6}$
- (10) $\Rightarrow \text{num} * \text{num} / \text{num}$

第二步 加入加、减运算

$\text{expr} \rightarrow \text{term rest5}$

$\text{rest5} \rightarrow +\text{term rest5} \mid -\text{term rest5} \mid \epsilon$

$\text{term} \rightarrow \text{unary rest6}$

$\text{rest6} \rightarrow * \text{unary rest6} \mid / \text{unary rest6} \mid \epsilon$

$\text{unary} \rightarrow \text{factor}$

$\text{factor} \rightarrow \text{num}$

输入:

9+5*2/3-6

输出 (二选一):

1) 按使用产生式过程

- (1) $\text{expr} \rightarrow \text{term rest5}$
- (2) $\text{term} \rightarrow \text{unary rest6}$
- (3) $\text{unary} \rightarrow \text{factor}$
- (4) $\text{factor} \rightarrow \text{num}$
- (5) $\text{rest6} \rightarrow \epsilon$
- (6) $\text{rest5} \rightarrow +\text{term rest5}$
- (7) $\text{term} \rightarrow \text{unary rest6}$
- (8) $\text{unary} \rightarrow \text{factor}$
- (9) $\text{factor} \rightarrow \text{num}$
- (10) $\text{rest6} \rightarrow * \text{unary rest6}$
- (11) $\text{unary} \rightarrow \text{factor}$
- (12) $\text{factor} \rightarrow \text{num}$
- (13) $\text{rest6} \rightarrow / \text{unary rest6}$
- (14) $\text{unary} \rightarrow \text{factor}$
- (15) $\text{factor} \rightarrow \text{num}$
- (16) $\text{rest6} \rightarrow \epsilon$
- (17) $\text{rest5} \rightarrow -\text{term rest5}$

- (18) $\text{term} \rightarrow \text{unary rest6}$
- (19) $\text{unary} \rightarrow \text{factor}$
- (20) $\text{factor} \rightarrow \text{num}$
- (21) $\text{rest6} \rightarrow \epsilon$
- (22) $\text{rest5} \rightarrow \epsilon$

2) 按推导过程

- (1) $\text{expr} \Rightarrow \text{term rest5}$
- (2) $\Rightarrow \text{unary rest6 rest5}$
- (3) $\Rightarrow \text{factor rest6 rest5}$
- (4) $\Rightarrow \text{num rest6 rest5}$
- (5) $\Rightarrow \text{num rest5}$
- (6) $\Rightarrow \text{num} + \text{term rest5}$
- (7) $\Rightarrow \text{num} + \text{unary rest6 rest5}$
- (8) $\Rightarrow \text{num} + \text{factor rest6 rest5}$
- (9) $\Rightarrow \text{num} + \text{num rest6 rest5}$
- (10) $\Rightarrow \text{num} + \text{num} * \text{unary rest6 rest5}$
- (11) $\Rightarrow \text{num} + \text{num} * \text{factor rest6 rest5}$
- (12) $\Rightarrow \text{num} + \text{num} * \text{num rest6 rest5}$
- (13) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{unary rest6 rest5}$
- (14) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{factor rest6 rest5}$
- (15) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num rest6 rest5}$
- (16) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num rest5}$
- (17) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{term rest5}$
- (18) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{unary rest6 rest5}$
- (19) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{factor rest6 rest5}$
- (20) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{num rest6 rest5}$
- (21) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{num rest5}$
- (22) $\Rightarrow \text{num} + \text{num} * \text{num} / \text{num} - \text{num}$